p. 252 If the meme is a scientific idea, its spread will depend on how acceptable it is to the population of individual scientists; a rough measure of its survival value could be obtained by counting the number of times it is referred to in successive years in scientific journals.

I'd hate it if this were taken to mean that 'catchiness' was the only criterion for acceptance of a scientific idea. After all, some scientific ideas are actually right, others wrong! Their rightness and wrongness can be tested; their logic can be dissected. They are really not like pop-tunes, religions, or punk hairdos. Nevertheless there is a sociology as well as a logic to science. Some bad scientific ideas can spread widely, at least for a while. And some good ideas lie dormant for years before finally catching on and colonizing scientific imaginations.

We can find a prime example of this dormancy followed by rampant propagation in one of the main ideas in this book, Hamilton's theory of

kin selection. I thought it would be a fitting case for trying out the idea of measuring meme spread by counting journal references. In the first edition I noted (p. 116) that 'His two papers of 1964 are among the most important contributions to social ethology ever written, and I have never been able to understand why they have been so neglected by ethologists (his name does not even appear in the index of two major text-books of ethology, both published in 1970). Fortunately there are recent signs of a revival of interest in his ideas.' I wrote that in 1976. Let us trace the course of that memic revival over the subsequent decade.

Science Citation Index is a rather strange publication in which one may look up any published paper and see tabulated, for a given year, the number of subsequent publications that have quoted it. It is intended as an aid to tracking down the literature on a given topic. University appointments committees have picked up the habit of using it as a rough and ready (too rough and too ready) way of comparing the scientific achievements of applicants for jobs. By counting the citations of Hamilton's papers, in each year since 1964, we can approximately track the progress of his ideas into the consciousness of biologists (Figure 1). The initial dormancy is very evident. Then it looks as though there is a dramatic upturn in interest in kin selection during the 1970s. If there is any point where the upward trend begins, it seems to be between 1973 and 1974. The upturn then gathers pace up to a peak in 1981, after which the annual rate of citation fluctuates irregularly about a plateau.



Figure 1. Yearly citations of Hamilton (1964) in the *Science Citation Index*

A memic myth has grown up that the upsurge of interest in kin selection was all triggered by books published in 1975 and 1976. The graph, with its upturn in 1974, seems to give the lie to this idea. On the contrary, the evidence could be used to support a very different hypothesis, namely that we are dealing with one of those ideas that was 'in the air', 'whose time had come'.